Phase 3: Actionable Intelligence

LoRA Fine-Tuning Qwen2.5-1.5B for JSON Entity Extraction

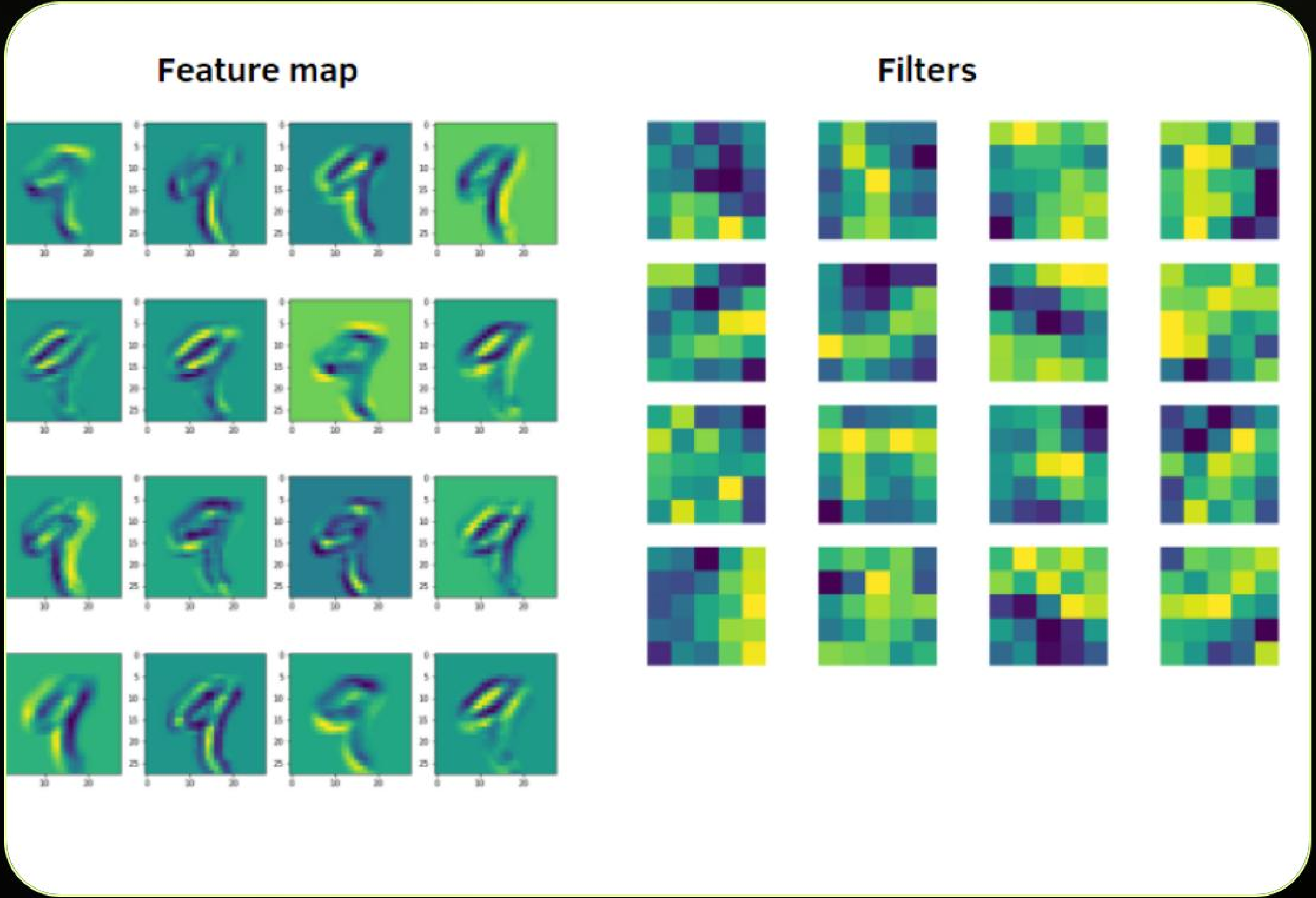
The Heavy Lifter

Bridging Intent to Action

The goal is to transform natural language into strict, machine-readable JSON payloads.

Qwen2.5-1.5B-Instruct provides a powerful foundation for reasoning and extraction, but requires specific formatting to prevent conversational "drift."

We leverage Parameter-Efficient Fine-Tuning (PEFT) to achieve high accuracy on a limited compute budget.



Efficiency via 4-Bit Quantization

4-Bit

NF4 Normal Float Precision

Memory Management

Using `bitsandbytes`, we quantize the 1.5B parameters into 4-bit precision.

This reduces memory overhead by nearly 4x, allowing the model to fit comfortably within the 16GB VRAM of a Colab T4 GPU during training.

Computation is handled in FP16 to maintain high accuracy and speed.

Parameter Efficiency: LoRA

Digital Data Flow

Low-Rank Adaptation

Instead of updating all 1.5 Billion weights, we inject small, trainable "adapter" matrices into the attention layers.

- Rank (r): 16 (Optimal for task complexity)
- Alpha: 32 (Scaling factor)
- Targeting: `q_proj`, `v_proj`, `k_proj`, `o_proj`

Result: Only training ~1-5% of total parameters.

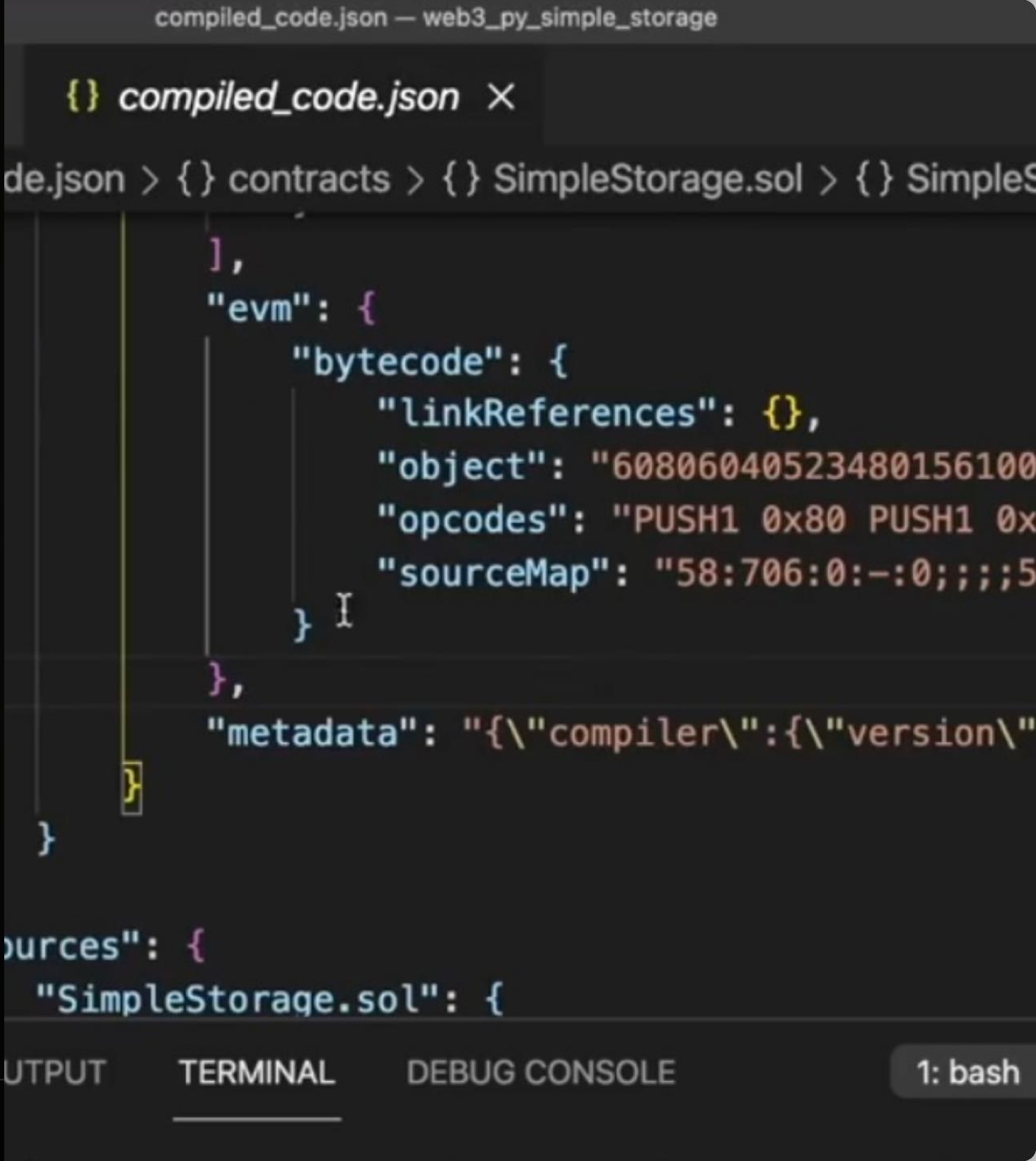
Structured JSONL Dataset

The ChatML Format

The model is trained on a specific conversational template to ensure deterministic outputs.

Input: User instruction containing order IDs or shipping addresses.

Target: Valid JSON object containing intent and extracted entities.



SFTTrainer Orchestration

Learning Rate

2e-4 with Linear Scheduler for stable convergence on small datasets.

Batch Size

Per device: 4. Grad Accumulation: 4. Effective batch size of 16.

Training Time

~1 Hour on T4 GPU for 3 Epochs / 1000 samples.

VRAM Optimization (Colab T4)

By using 4-bit quantization, we leave ample buffer for the optimizer and gradient accumulation.

The Pipeline Transformation

Zero Conversational Fluff

Standard LLMs often preface answers with "Sure, I can help you with that!"

Our fine-tuning forces the model to respond **only** with the JSON string, making it safe for `json.loads()` in the backend.

f (Input) = JSON

Inference Validation

Input

"I want to change my shipping address to 123 Main St for order #998877."

Extraction

Intent: change_shipping_address
Order_id: 998877
Address: 123 Main St

Result


```
{"intent":
"change_shipping_address",
"order_id": "998877",
"new_address": "123 Main St"}
```

Weight Preservation

- **Adapter Only:** We save only the LoRA weights (~50MB), not the entire base model.
- **Model Fusion:** At runtime, the adapter is merged into the base weights for inference.
- **Versioning:** Export to Hugging Face Hub for direct integration with the FastAPI production server.

The Final Milestones

Fine-Tune

Run training script on Colab T4.

Validate

Test with complex actionable prompts.

Export

Save LoRA weights to cloud storage.

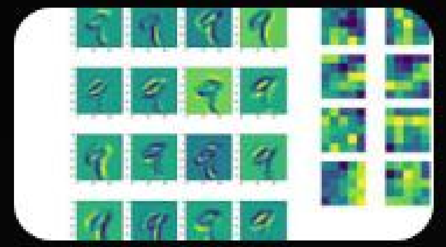
Deploy

Connect to FastAPI and Gradio UI.

Questions?

Phase 3 Implementation: Actionable LLM Path Completed

Image Sources



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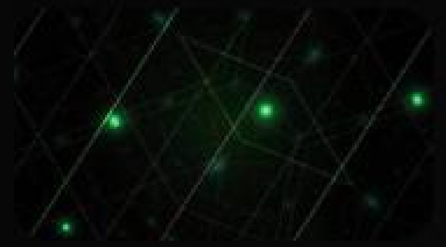
Source: medium.com



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